

Samuel Wang

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EDUCATION

McMaster University

Bachelor of Engineering in Electrical and Biomedical Engineering

Hamilton, ON

Sep. 2023 – June 2028

EXPERIENCE

McMaster University Aerial Robotics and Drones Team

Electrical Sub-team Co-Lead

Hamilton, ON

Feb. 2026 – Jun. 2026

- Promoted due to exceptional leadership ability and technical experience
- Mapped electrical infrastructure of 12S (44.4V) quadcopter drone, including power and communications systems

Electrical Subteam Member

Mar. 2025 – Feb. 2026

- Optimized embedded system circuitry using Altium Designer, LTSPICE simulation, and circuit analysis
- Conducted error detection and peer-to-peer analysis of power systems, improving battery life by 25 percent
- Developed guidelines and best practices, organized meetings, and provided technical guidance to a 6-person team

Canadian Imperial Bank of Commerce (CIBC)

Quality Assurance, Quality Engineering, and Automation Co-op Student

Toronto, ON

May 2024 – Aug. 2024

- Developed Python scripts to fetch data from JIRA using SQL queries based on daily user-provided filters as part of team project to create centralized, company-wide Quality Assurance metrics dashboard under Agile methodology
- Processed and validated JIRA data for storage in PostgreSQL database using Python
- Automated formation and validation of data reports in Microsoft Excel using Python
- Nominated by manager to the CIBC SPARK student leadership program for collaboration and dedication
- Attended weekly team meetings with clients and QE leadership team to maintain strong communication

PROJECTS

Modular Robotic Arm | *Python, C++, Autodesk Inventor, Raspberry Pi, KiCAD, Linux*

Mar. 2026 – Present

- Personal enrichment project with focus on embedded systems and software processing
- Designed 6-joint, modular robotic arm with 360° of motion using Autodesk Inventor
- Developing embedded systems in KiCAD for 11.1V LiPo battery monitoring, power distribution, and protection
- Creating PID servo motor controller and software filters in C++ and Python using position and current feedback

Handheld LiDAR Scanner | *C++, MATLAB, TI MSP-EXP432 Microcontroller*

Jan. 2026 – Apr. 2026

- Developed various embedded systems for a handheld LiDAR system with a 4m range
- Created 360° rotating LiDAR scanner, LED troubleshooting systems, and control unit embedded hardware systems
- Used C++ to program MSP-432EXP4E401Y microcontroller in the Keil uVision IDE with controllable scan / upload modes, 7 automatic error detection modes, and automatic range switching
- Created collection of three MATLAB scripts to map and store data as 3D surface maps

LiDAR Mounting Case | *Autodesk Inventor*

Mar. 2026 – Apr. 2026

- Designed a 3D-printable components case to mount an MSP-EXP432E401Y microcontroller and LIDAR system
- Optimized design for simplicity and compactness for sub-5 hour printing time on PrusaSlicer (25 x 18.5 x 3.6 cm)
- Successfully partnered with McMaster COMP ENG 2DX3 professors to provide cases for over 100 future students

TECHNICAL SKILLS

Software: Autodesk Inventor (5+ years), Python (5+ years), MATLAB (4 years), C++ (3 years), Altium Designer (1 year), PSPICE (1 year), Keil uVision (1 year), GitHub (1 year), Assembly (<1 year), JIRA (<1 year), PostgreSQL (<1 year), KiCAD (<1 year), Linux (<1 year), HTML/HTML5 (<1 year), JavaScript (<1 year), CSS (<1 year)

Languages: English (Fluent), French (Conversational), Mandarin (Conversational)

Laboratory: Circuit construction, Soldering, Troubleshooting and software/hardware debugging, Oscilloscope and multimeter measurements, Cell culture cultivation, Scratch assays, Microscope analysis, PCR procedure, Gel electrophoresis assay, Western blot assay, Biosafety cabinet (BSC) work, Micro pipetting, WHMIS safety procedure